

Soil Acidity in Tanzania:

Recommendations for Targeting of
Remediation Efforts at Scale

1. PURPOSE OF THIS TECHNICAL NOTE

Significant portions of Tanzania's cropland are affected by regional soil information systems – such as TanSIS in Tanzania. Soil acidity has reemerged as a priority topic for agricultural development, with some discussion around the potential mobilization of investments to address areas affected by acid soils. To support and guide these efforts, the International Wheat Improvement Center (CIMMYT), under the Guiding Action for Soil Health by the Bill and Melinda Gates Foundation. The purpose of this technical note is on acid soil management and crop yield response to lime in Tanzania. The brief draws on analysis of lime trials data and a broader-scale spatial analysis designed to inform targeting.

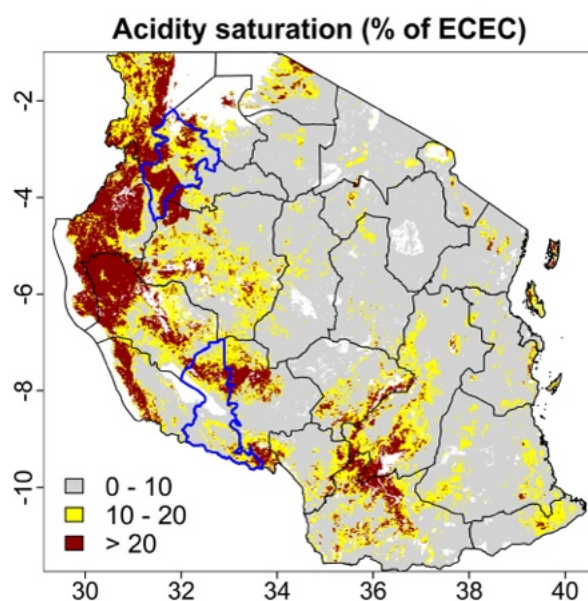
2. SPATIAL DISTRIBUTION OF SOIL ACIDITY IN TANZANIA

In recent years, mapped estimates of areas with pH below 5.5 in Tanzania, which is consistent with targeting approaches where soil acidity is a key approach. This reflects the typical association of low pH soils with low nutrient uptake by crops. Exchangeable acidity can be used as a proxy for soil acidity. Analysis of Tanzanian soils shows that many low pH areas are in the central and southern regions. The core problem is more related to poor soil fertility with low organic matter management combining mineral and organic fertilizer input.

For targeting the promotion of agricultural lime, we recommend targeting areas with ECEC), as these are the areas most likely to experience high exchangeable acidity criteria.¹ There are rates of up to 2 t/ha would be recommended.

However, in cropland with low levels of exchangeable acidity, other alternatives are available, namely integrated soil fertility management. On-going analysis should clarify this with results expected in 2025, but for now we recommend

Figure 1: Distribution of exchangeable acidity, as a % of the effective cation exchange capacity (ECEC), across Tanzania



Source: <https://www.acidsoils.africa/>

3. LIME REQUIREMENTS TO REMEDIATE ACID SOILS

It is important to target investment resources for large pH and high levels of exchangeable acidity. Our analysis estimated 2.4 million metric tonnes of lime to bring exchangeable acidity levels to <10%. There are minimal yield penalties for most crops.

Lime recommendation (to treat soil acidity) depends on acidity saturation. To reduce acidity saturation to <10%, the recommended lime application rates vary. The variation in recommended rates: 32% of the cropland would require lime rates above 1 t/ha, and the remaining 68% would require lime rates not harmful for crop production. It is also important to consider the respective lime rates. For cereals, 70, 20 and 10% of 1 t/ha, respectively. For legumes, 45, 30 and 25% of the 1 t/ha, respectively.

Localized lime recommendations can vary substantially with soil type and should be defined at more localized levels. For field specific soil testing efforts, focused on exchangeable acidity, a more detailed analysis is needed.

4. YIELD RESPONSE TO LIME AND ASSOCIATED PROFITABILITY

Agronomy research evaluated the impact of different rates of lime application on about 100 farmers' fields over 2 growing seasons. The results were from the Geita and Mbozi in the Southern highlands.

The return from liming varies across application rates. At rates of 1 t/ha for maize and up to 2.5 t/ha for bean, the return is not cost effective at current prices of lime and crop output. The return is not cost effective at current prices of lime and crop output in most crops and districts. Close to 1 t/ha lime leads to first profitable returns for maize in Mbozi. However, while equating yields in both Geita and Mbozi, the magnitude of the profit from lime is not profitable for bean in either location.

The results show that even where crop responses to liming are profitable especially at high lime rates. Furthermore, high lime rates are not cost effective due to soil related constraints and should only be prioritized where the response and profitability is high. Lime application rates of 1 t/ha or more are not cost effective in such areas alternatives to lime (such as improved crop varieties).

Given the large amounts of lime that must be moved over long distances, the important constraints are the factory-gate cost of lime and transportation costs, not just on extraction and production but also on transport. Even if new sources of lime are developed domestically, the cost of lime is not profitable at current market prices. Even if the Government subsidizes lime usage improved, the aggregate returns to lime are not profitable in areas where the unsubsidized farm-level cost-benefit ratio is less than 1.

5. RECOMMENDATIONS

Recommendations for liming investment in Tanzania have been different parts of the country challenged with soil acidity targeting of remediation efforts at scale across all the country on targeting lime distribution and promotion efforts, policy Tanzania.

The recommendations for targeting lime distribution and promotion targeting lime recommendation rates to local conditions; (b) prioritizing legume, root and tuber over cereal crops; (d) data, promote modest lime rates of 1-2 t/ha to avoid risks

Policy related recommendations include (i) strengthening awareness, knowledge, and learning on use of lime; (ii) appropriate targeting per recommendations, and (iii) consi